Which Parts of BM25 Help Learned Top-k Selection? Tests in Sparse Retrieval, KV-Cache Page Selection and Expert Routing

Team ANLP Doomsday

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IIIT Hyderabad, India https://github.com/avi1o1/ANLP-Doomsday

Abstract

BM25 corrects a term-match score with inverse document frequency (IDF), saturation and length normalisation. We test which parts help three learned top-k selectors that use none of them, in sparse retrieval, KV-cache page selection and mixture-of-experts routing. On thirteen collections, with fourteen sparse codes from one dense encoder, the full correction raises mean nDCG@10 from 0.036 to 0.158. Saturation, which needs no collection statistics, gives 62% of this gain and IDF 10%, and in tests fixed in advance the spread of a code's stored values ranks saturation's effect. The Zipf-like frequency curve credited by prior work does not predict the gain, and agreement with text BM25 is not a reliable label-free test of whether IDF and length normalisation help. In KV selection and routing, where an item's frequency is its own selection rate, saturation and length change mean accuracy by at most 0.02, but IDF lowers mean RULER accuracy below 0.05 on two models and OLMoE's MMLU accuracy by 0.09 to 0.10.

1 Introduction

BM25 scores a document by its term matches with the query, with three corrections to the raw score (Spärck Jones, 1972; Robertson and Zaragoza, 2009). Inverse document frequency (IDF) gives little weight to a term that occurs in most documents. Saturation limits the reward for repetition. In a document of average length, with BM25's usual $k_1 = 1.2$, a term that occurs ten times scores only a quarter more than one that occurs three times. Length normalisation discounts long documents, which match more terms by chance.

Three learned components of NLP systems keep the top k items by a score and use none of these corrections. A learned sparse retriever scores a document by an inner product of non-negative query and document features and returns the top k (Formal et al., 2021b; Nguyen et al., 2023). Quest, a sparse-attention method, scores fixed-size pages of the key-value (KV) cache against the current query and attends only to the highest-scoring pages (Tang et al., 2024). A mixture-of-experts router scores every expert for a token and sends the token to the top k (Shazeer et al., 2017; Fedus et al., 2022). In each the score is non-negative or can be made so, and each item can be given a frequency, so BM25's correction applies to all three (Figure 1).

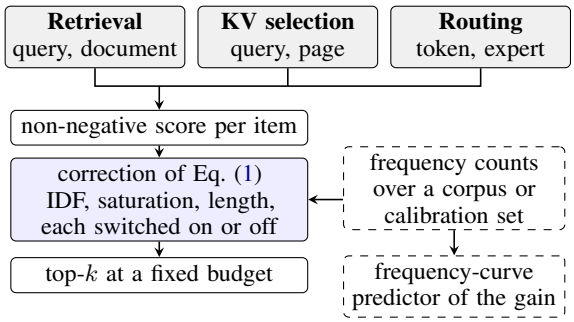


Figure 1: The three settings. Each scores items against an input and keeps the top k. The score sums features in retrieval and is one value per page or expert. The BM25 correction is applied between scoring and top-k selection. Its frequency counts are also the input to the frequency-curve predictor of its gain.

Applied to learned features, BM25 has both helped and hurt. On one benchmark suite it beats the inner product by 0.20 nDCG@10 on sparse-autoencoder features and loses by 0.11 on SPLADE-v3 (Clavié et al., 2026). The explanation offered is that BM25 helps when the features' document frequencies follow a Zipf-like curve, as word frequencies do (Clavié et al., 2026; Han et al., 2026). If that is right, statistics of the frequency curve should predict the sign of the gain before the experiment is run.

We test which parts of the correction help, and whether the gain can be predicted before it is measured, in all three settings. In retrieval we hold the encoder fixed and vary only the sparse code built on

it. Fourteen codes come from one dense encoder, multilingual E5-base, and each is scored on thirteen collections with each part switched on or off, eight configurations in all. Exact Shapley values split each gain among the parts. A frequency-curve predictor estimates the gain, and a label-free rule predicts from agreement with a reference ranker whether IDF and length normalisation help. We fixed the main tests, and any pass marks, before reading their results, and call them registered (Table 4 in the appendix); the other analyses are exploratory.

1. The correction raises the E5 codes' mean nDCG@10 over the 182 code–collection pairs from 0.036 to 0.158. On six collections the best code under the correction beats text BM25 (Section 5.1).
2. Saturation gives most of the gain (62%) and IDF the least (10%). Saturation needs no collection statistics, and in registered tests the spread of the values stored in the index ranks the size of its effect at held-out values of k_1 (Sections 5.2 and 5.3).
3. Statistics of the frequency curve do not predict the gain (Section 5.4). Fitted models that use agreement with text BM25 explain about half the held-out variance of the gain from IDF and length normalisation, but the label-free rule based on that agreement is no better in sign than a constant on 84 later rows and fails its registered test on SPLADE-v3 (Sections 5.5 and 6).
4. In KV selection and routing, with frequency counted from the selector's own choices, IDF lowers RULER accuracy to near zero on two models and MMLU accuracy by 0.09 to 0.10, and the frequency-curve predictor fails every registered test. With IDF off, saturation provably cannot change these selections, up to ties. In KV selection, exact attention mass shows the harm of IDF without labels. The registered rule built on it matches an oracle but fails its sign test on both models, because adding IDF and length helps on no row and a constant does as well (Sections 3.4 and 7).

2 Related Work

Term weighting and learned sparse retrieval. IDF began as a statistical account of term specificity (Spärck Jones, 1972). Saturation approximates the two-Poisson model of within-document frequency (Robertson and Walker, 1994), and pivoted length normalisation keeps long documents from being favoured or penalised (Singhal et al., 1996). BM25 combines the three (Robertson and Zaragoza, 2009). Learned sparse retrievers learn term weights through contextual exact match (Gao et al., 2021) or vocabulary expansion (Formal et al., 2021b,a; Lassance et al., 2024).

BM25 on learned features. Sparse autoencoders (SAEs) decompose dense activations into many non-negative features (Huben et al., 2024; Gao et al., 2025). Clavié et al. (2026) score SAE features of a frozen dense retriever with BM25 and report 0.582 nDCG@10 against 0.381 for the inner product, and the reverse on SPLADE-v3, 0.455 against 0.568. CL-SR loses 11 to 18% of its nDCG@10 on TREC-DL without IDF (Park et al., 2025). Han et al. (2026) find BM25 and the inner product about equal on image features and credit top-k filtering for suppressing what IDF would down-weight. None of these varies the representation systematically or separates all three parts. We hold the encoder fixed, derive fourteen codes from it and switch each part on and off, so the gain can be split among the parts and set against properties of each code.

Predicting effectiveness without labels. Query performance prediction estimates retrieval quality from the query and the ranking (Carmel and Yom-Tov, 2010; Cronen-Townsend et al., 2002; Shtok et al., 2012), and systems can be ranked against pseudo-relevant documents (Soboroff et al., 2001). Our target, the difference between two scorers, is new.

Selection inside the transformer. Quest scores KV pages against page min-max key summaries (Tang et al., 2024). H2O keeps recent tokens and heavy hitters, the tokens that have received the most attention (Zhang et al., 2023b). SnapKV keeps the positions recent queries attend to (Li et al., 2024), StreamingLLM the first tokens and a recent window (Xiao et al., 2024), and Expected Attention the positions future queries are predicted to attend to (Devoto et al., 2025). Mixture-of-experts layers route each token to its top-k experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022), balanced by an auxiliary loss (Zoph et al., 2022) or a load-driven bias (Wang et al., 2024a), whose analogue we run as a baseline. H2O's heavy-hitter rule is the opposite of IDF when frequency is a selection rate. A load-driven bias moves tokens

towards under-used experts, the same direction as IDF, but adds a term to the router's score instead of multiplying it. We know of no work that applies BM25's parts to KV selection or routing.

3 Method

3.1 BM25's correction as three switches

For query and document vectors x_q and x_d with non-negative coordinates (features), we score

$$S(q,d) = \sum_j x_{qj}\, w_j\, T(x_{dj}, \ell_d), \qquad (1)$$

and call x_{dj} the stored value of feature j in document d. Each part can be switched on or off. With IDF on, the weight w_j is $\mathrm{idf}_j = \ln(1 + (N - df_j + 0.5)/(df_j + 0.5))$, where feature j is nonzero in df_j of the N documents, and with IDF off $w_j = 1$. With saturation on, the transform is $T(f,\ell) = (k_1+1)f/(f + k_1\ell)$, and with it off $T(f,\ell) = f/\ell$. With length normalisation on, $\ell_d = 1 - b + bL_d/\overline{L}$, where $L_d = \sum_j x_{dj}$ is the document's length and $\overline{L}$ its mean over the collection, and with it off $\ell_d = 1$. We use BM25's usual $k_1 = 1.2$ and $b = 0.75$ unless stated.

We write a configuration as three bits in the order IDF, saturation, length. Configuration `000` (all-off) is the plain inner product, `111` (all-on) is BM25, and `010` is saturation alone. IDF and length normalisation need collection statistics (document frequencies and mean length), while saturation uses only the document's own values.

3.2 Measuring each part's share of the gain

A row is one sparse code on one collection at one setting of the support budget (a cap on the number of nonzero features per query and per document), k_1, b and the seed used to fit the code. It carries eight nDCG@10 values Q_c, one per configuration c. The 182 primary rows are the fourteen codes on the thirteen collections at the default settings.

The gain of the full correction is $Q_{111} - Q_{000}$, and we split it among the parts by Shapley values (Shapley, 1953). The Shapley value ϕ_s of part s is the change in nDCG@10 when s is switched on, averaged over the six orders in which the three parts can be switched on. Writing Q_A for the configuration with the parts in A on,

$$\phi_s = \sum_{A \not\ni s} \frac{|A|!\,(2-|A|)!}{6} \big(Q_{A\cup\{s\}} - Q_A\big), \qquad (2)$$

and the three values sum to the gain.

3.3 Predicting the gain without measuring it

We predict two quantities. The frequency-curve predictor predicts the full gain, scaled as the margin m below. The label-free rule predicts the sign of $G = Q_{111} - Q_{010}$, the gain from adding IDF and length normalisation once saturation is on. The sign of G thus decides whether collection statistics are worth computing.

The frequency-curve predictor. Our project proposal planned a predictor of the gain from the shape of the feature-frequency curve, as prior work implies, and we fixed it before any retrieval result existed. It is a ridge regression on three statistics of the curve (the Zipf slope, the variance of IDF and the share of activations in the most frequent 1% of features). Its target is the margin $m = (Q_{111} - Q_{000})/\operatorname{sd}(Q_{000}, \ldots, Q_{111})$, the gain divided by the spread of the eight scores, so that rows of different difficulty share one scale. The margin is undefined in the three primary rows where all eight scores are equal.

The label-free rule. Without relevance labels, each configuration's ranking can still be compared with a reference ranker. We count the reference's top 10 for each query as relevant and score each configuration against these pseudo-labels by nDCG@10, giving a pseudo-quality $\tilde{Q}_c$. Putting $\tilde{Q}$ in place of Q gives an estimate $\tilde{G}$ of G with no fitted parameter. The rule uses `111` where $\tilde{G} > 0$ and `010` otherwise. In retrieval the registered reference is text BM25, and the dense encoder E5, alone or fused with text BM25 by reciprocal rank, is a secondary reference. In KV selection $\tilde{Q}_c$ is the exact attention mass of c's pages (Section 4), with token F1 to the full-attention output as a secondary measure.

Pass marks. The label-free rule passes if its sign is right on more rows than the better constant (the better of always predicting a gain and always predicting a loss) and its choices score at least as well as all-on on average. We also compare it with always choosing `010` and with an oracle that picks the better of `111` and `010` in each row. Outside retrieval the frequency-curve predictor must get the sign right on at least 0.70 of rows, keep its mean absolute error (MAE) on m within 1.81 (1.5 times its root mean squared error in retrieval), and beat always predicting a gain, the majority sign in retrieval.

3.4 Applying the correction to KV pages and experts

Eq. (1) applies to any selection with a non-negative score per item and a frequency count. A KV page or an expert carries one score, so the sum has a single term.

KV pages. Each page i of 16 cached tokens in an attention head is one item. To count frequencies, we run Quest at 64 positions of the example's own prompt (calibration). A page's value f_i is Quest's criticality, an upper bound on the query–key product from the page's key minima and maxima, passed through a softplus and divided by the head's median selected value during calibration. For page i, N counts the calibration positions at or after it, and df_i those at which Quest selects it. The page's length is the norm of its concatenated key minima and maxima.

Experts. For expert e the value is the router probability, the frequency is the share of calibration tokens routed to e, and the length is the norm of the token's router logits, which all experts of a token share. The selected experts keep their original gate weights.

In both settings an item's frequency is the selector's own selection rate, so IDF gives low weight to the items the selector picks often, and IDF's effect here partly follows from this definition.

Proposition 1. *When each item carries a single score $f_i \geq 0$, as a KV page or an expert does, and IDF is off, the saturated score is a strictly increasing function of f_i/ℓ_i and selects the same top k as the unsaturated score, up to ties.*

It holds because $(k_1+1)f/(f+k_1\ell) = g(f/\ell)$ with $g(f) = (k_1+1)f/(f+k_1)$, and g is strictly increasing on $f \geq 0$. With IDF on, $\mathrm{idf}_i\, g(f_i)$ and $\mathrm{idf}_i f_i$ can order items differently. In KV selection pages differ in length, so `001` can differ from `000`, but `010` selects as `000` does and `011` as `001`. In routing all experts of a token share one length, so the four configurations with IDF off route identically up to ties, as do `100` and `101`.

A row is one RULER task at one token budget in KV selection, and WikiText or one MMLU subject in routing. The frequency-curve predictor is applied without refitting.

4 Experimental Setup

Sparse codes. All retrieval codes come from `multilingual-e5-base` (E5, Wang et al., 2024b) at a pinned revision. Seven bases map an E5 state to a non-negative sparse vector. They are the identity, a Gaussian projection to 768 dimensions and PCA to 384, each split into positive and negative parts; soft k-means over $4{,}096$ centroids; product and residual quantisation (PQ, RQ; 8×256, Jégou et al., 2011; Chen et al., 2010); and a Top-K SAE with $8{,}192$ features of which at most 32 are active. All but the identity are fitted on the same $200{,}000$ MS MARCO passages (Bajaj et al., 2016). The bases form four families for held-out validation (linear, clustered, quantised and learned). A basis is applied either to the mean of the token states (pooled) or to each token state before summing (token), and each basis at one granularity is one sparse code. The primary support budget keeps at most 32 nonzero features per query and 128 per document, and document frequencies, lengths and IDF come from these capped vectors in each corpus. Retrieval is exact over the whole corpus.

Collections and sweeps. The thirteen collections come from eleven corpora. They are MS MARCO dev with TREC-DL 2019 and 2020 on the same documents, six BEIR sets (Thakur et al., 2021), LIMIT (Weller et al., 2026) and the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a). Beyond the primary rows, sweeps vary one setting each (Table 7 in the appendix). SPLADE-v3 (Lassance et al., 2024), with its own backbone, is scored under the same eight configurations as an external comparison.

Significance. All-on is compared with all-off by a paired bootstrap of $Q_{111} - Q_{000}$ over queries or examples ($10{,}000$ draws; $2{,}000$ for SPLADE-v3 and the transfer settings), with Holm's correction within each group of rows declared in advance (Holm, 1979).

KV-page selection. The attention model is Qwen2.5-1.5B-Instruct (Qwen Team, 2024) in float16, chosen before any run, and Llama-3.2-1B-Instruct is run with the same protocol as a registered replication. All 13 tasks of the RULER long-context benchmark (Hsieh et al., 2024), from needle retrieval to QA, are run at a 4K context with 100 examples per task and token budgets of 512 and 1,024 over 16-token pages. Prefill is dense and decoding attends only to the selected pages. Full attention, Quest and H2O (Zhang et al., 2023b) are run beside the eight configurations. Configuration `000` is Quest's scoring, and the softplus and per-

head scaling keep Quest's order. The label-free reference is exact attention mass, the share of full attention's softmax mass that falls on the pages a configuration would select, measured along the full-attention generation.

Expert routing. OLMoE-1B-7B (Muennighoff et al., 2025) routes each token to 8 of 64 experts. Quality is the mean token log-likelihood on WikiText-2 (Merity et al., 2017) in 2,048-token windows, reported as perplexity, and zero-shot accuracy on each of the 57 MMLU subjects (Hendrycks et al., 2021), scored by the likelihood of each choice. Frequencies come from 256 training-split windows for WikiText and from each subject's own prompts for MMLU. A load-driven frequency bias, the analogue of loss-free balancing (Wang et al., 2024a), is run beside the eight configurations.

5 Results in Retrieval

5.1 The correction raises the codes' nDCG@10

Under the inner product the sparse codes are weak. Their mean nDCG@10 is at most 0.135 on every collection (Table 1) and 0.036 over the 182 primary rows. The full correction raises it on every collection, to 0.158 over the primary rows. All-on beats all-off in 162 of the 182 primary rows, significantly after Holm correction in 134. On six collections the best code under all-on beats text BM25, three of them on the MS MARCO corpus, two from MIRACL, and FiQA. The best code is the token-level SAE on every collection but LIMIT. No code beats dense E5 except on LIMIT, where E5 scores 0.015 and only token-level k-means exceeds it (0.033).

In 81 rows the inner product scores below 0.01, so nearly any correction beats it. All-on is also not the best configuration. It is the strict best of the eight in only 42 rows, and `011` (saturation and length without IDF) has a higher mean, 0.163 against 0.158, though not at the 16/64 budget (0.101 against 0.104).

5.2 Saturation gives most of the gain

Saturation has the largest Shapley value at every support budget (Table 2). Over the 182 primary rows it gives 62% of the gain, length normalisation 28% and IDF, the part prior work credits, 10%. The two parts that need collection statistics give 38% (95% interval 36 to 41%, resampling collections). Saturation alone (`010`) raises mean nDCG@10 from 0.036 to 0.137. As the budget grows, IDF's share shrinks and length's grows, and with uncapped (native) support IDF's mean Shapley value is slightly negative. Han et al. (2026) report the same trend for image features, where document frequencies become more alike as more features are active. Shuffling the frequencies turns IDF's Shapley value negative, but removing IDF from all-on does not lower mean nDCG@10, over the primary rows (`011` scores 0.163) or with uniform frequencies on three collections (Appendix A).

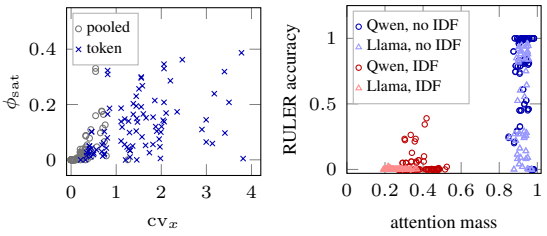


Figure 2: Left, the spread of the stored values against saturation's Shapley value, one point per primary retrieval row. Right, the share of full attention's mass on the selected pages against RULER accuracy, one point per KV row and configuration with a mass reference.

5.3 The spread of the stored values predicts saturation's effect

With length normalisation off, saturation maps a stored value f to $g(f) = (k_1 + 1)f/(f + k_1)$, and the ratio $g(f)/f$ falls as f grows, so large values are shrunk more than small ones. If every stored value in an index were equal, g would multiply every score by one factor and the ranking would not change. Pooled PQ and RQ are such codes, with one active feature of value one per sub-quantiser or stage, and their saturation and length effects are exactly zero in every row (Table 5 in the appendix). The more unequal the values, the more saturation can reorder a ranking. We measure this inequality by the coefficient of variation cv_x of the nonzero values stored in the index. In our codes the reordering usually helps ($\phi_{\text{sat}} > 0$ in 79% of primary rows), and it grows with cv_x (Spearman 0.78, Figure 2, left).

We fixed six hypotheses on this mechanism in advance and tested them on the k_1 and b sweep of five collections, mostly new rows (Table 6 in the appendix). All six pass. H2 predicts that a smaller k_1 gives a larger ϕ_{sat}, H3 that cv_x and ϕ_{sat} correlate within a corpus, and H6 that a larger b gives a larger length effect. H5 (no length effect at $b = 0$) checks the implementation, and H1 and

	Baselines		E5 sparse codes			SPLADE-v3	
Collection	BM25	E5	000	111	best	000	111
MS MARCO dev	.222	.425	.013	.130	.318	.470	.382
TREC-DL 19	.456	.704	.036	.255	.604	.722	.680
TREC-DL 20	.484	.697	.035	.251	.556	.751	.674
ArguAna	.411	.444	.021	.130	.373	.489	.563
FiQA	.236	.381	.013	.084	.265	.380	.318
NFCorpus	.307	.326	.064	.115	.286	.364	.356
SCIDOCS	.152	.172	.009	.044	.121	.156	.154
SciFact	.662	.694	.088	.197	.426	.717	.716
TREC-COVID	.582	.695	.135	.246	.514	.731	.583
LIMIT	.939	.015	.001	.004	.033	.447	.667
MIRACL hi	.357	.582	.011	.124	.329	.008	.004
MIRACL bn	.407	.700	.025	.217	.488	.000	.001
MIRACL te	.331	.751	.019	.259	.582	.001	.001

Table 1: nDCG@10 of text BM25 and of E5, the dense encoder the codes come from; the mean over the fourteen codes under 000 and 111; the best single code under 111; and SPLADE-v3 under both. MIRACL is outside SPLADE-v3's vocabulary.

Setting	Rows	IDF	Sat.	Len.	Sum
E5 codes, primary	182	.012	.075	.034	.122
E5 codes, 16/64	182	.010	.045	.018	.074
E5 codes, 64/256	182	.013	.095	.056	.163
E5 codes, native	42	−.002	.076	.066	.140
SPLADE-v3	10	.011	.048	−.073	−.013
KV pages, Qwen	24	−.775	−.003	.002	−.777
KV pages, Llama	23	−.509	.000	.006	−.504
experts, MMLU	57	−.094	−.004	−.001	−.099
experts, WikiText	1	−.050	−.002	.000	−.051

Table 2: Mean Shapley value of each part (Eq. (2)) and their sum $Q_{111} - Q_{000}$, in each setting's metric (nDCG@10 in retrieval, accuracy in KV selection and MMLU, log-likelihood per token on WikiText). 16/64 and 64/256 are the query and document support caps of the other budgets, and native is uncapped.

one arm of H6 reuse primary rows. In H4 a ridge model of ϕ_{sat} on cv_x fitted at $k_1 = 1.2$ ranks the rows by ϕ_{sat} at $k_1 = 0.6$ and 2.4 (Spearman 0.87, 0.90).

5.4 The frequency curve does not predict the gain

The frequency-curve predictor explains almost none of the held-out variance of m, with R^2 of 0.04 when a basis family is held out and 0.02 when a corpus is held out. Its sign is right on 0.90 of rows, no more often than always predicting a gain. Part of the failure lies in the target. Of the margins, 85% fall between 1.5 and 3.5, because in many rows Q_{000} is near zero and every other configuration is far above it, so m reflects that pattern more than the size of the gain.

We also fitted exploratory ridge models of each Shapley value and of G directly, on three nested feature sets (Table 8 in the appendix). The first holds the three frequency-curve statistics. The second adds index statistics, which describe the nonzero values stored in the index, chiefly cv_x. The third adds the pseudo-qualities of Section 3.3 and features of each configuration's top-100 ranking. Index statistics predict saturation's Shapley value (held-out R^2 of 0.29 to 0.56) but predict G no better than the frequency curve does (0.01 to 0.12 against 0.01 to 0.13). G varies even within a code. For token-level SAE and PCA, the two strongest codes, $G > 0$ on 6 of 13 collections.

5.5 Agreement with text BM25 in place of labels

With pseudo-labels from text BM25 added, the fitted models reach positive held-out R^2 on every target and every split, and for G they reach 0.44 to 0.59, against at most 0.13 without pseudo-labels (Table 8). Within a row the pseudo-qualities order the eight configurations almost as the true scores do (median Spearman 0.98). The label-free rule was designed on the primary rows. On the 181 of them used in the fitted models, all-on averages 0.1590, the rule 0.1628 and the oracle 0.1654, so the rule recovers 59% of the gap to the oracle.

Our interim report stated the rule, unregistered, before we produced 84 new rows, the fourteen codes at the 16/64 and 64/256 budgets on the three MS MARCO collections. All-on and the oracle are close on these rows (0.202 and 0.203). With the text BM25 reference, $\tilde{G}$ tracks G (Spearman 0.91), but its sign is right on 0.87 of rows, against 0.88 for always predicting a gain. The dense or fused

reference gets the sign right on 0.95, but the dense reference is the encoder the codes come from. The three collections share one corpus, so these rows are not independent.

6 The Correction on SPLADE-v3

On SPLADE-v3 the correction costs a little, all of it through length normalisation, and the label-free rule with the text BM25 reference fails. Mean nDCG@10 falls from 0.523 under the inner product to 0.509 under all-on, and falls on 8 of 10 collections (Table 1). All-on is significantly worse on 4 collections and significantly better on ArguAna and LIMIT (bootstrap $p < 0.05$, uncorrected). The direction matches the reversal of Clavié et al. (2026); the size, 0.013 against their 0.113, does not. Without length normalisation, IDF and saturation (`110`) score 0.528, above the inner product (Appendix A).

The SPLADE-v3 scores were known before this test was fixed, but the label-free estimates were not. The sign of G splits 5 to 5, so a constant sign scores only 0.50. With the registered text BM25 reference, $\tilde{G}$ has the right sign on 5 of 10 collections and Spearman -0.04 with G. On the three MS MARCO collections, adding IDF and length normalisation raises agreement with text BM25 by 0.06 to 0.08 while lowering nDCG@10. Configurations that behave like BM25 agree more with text BM25, a bias noted in advance. The secondary dense reference meets both pass marks. Its sign is right on 7 of 10 collections, and its rule scores 0.519, against 0.509 for all-on and 0.515 for `010`. Over 10 collections the 95% interval for its gain over all-on includes zero. The fused reference fails.

7 Transfer to KV Pages and Experts

7.1 KV-page selection

Without IDF the corrected selectors match Quest, and with IDF they fail. On Qwen the four configurations without IDF stay within 0.02 of Quest's mean accuracy (0.71 and 0.78 at the two budgets), and the four with IDF score 0.015 to 0.046 (Table 3). Variable tracking, run without RULER's answer prefix, scores zero on Qwen, so its two rows have no defined margin. After Holm correction all-on is significantly worse than all-off in all 24 remaining rows and better in none, and G is negative in all 24. The mean Shapley value of IDF is -0.775, against -0.003 for saturation and 0.002 for length (Table 2).

The replication on Llama-3.2-1B-Instruct shows the same pattern. With IDF on, mean accuracy is at most 0.002, against 0.43 and 0.46 for Quest, and without IDF it stays within 0.02 of Quest. Three rows score zero in every configuration. In the other 23, all-on is significantly worse than all-off in 18 after Holm correction and better in none, and IDF's mean Shapley value is -0.509.

IDF gives low weights to the pages Quest selects often. A page selected at all 64 calibration positions gets the smoothed IDF $\ln(1 + 0.5/64.5) \approx 0.008$, and one never selected gets $\ln 130 \approx 4.9$. On the first single-needle task with Qwen at the 1,024 budget, the page holding the needle ranks between second and tenth of 210 to 247 pages by selection rate, so IDF gives it a low weight. Averaged over tasks, the pages a configuration with IDF would select hold 0.29 to 0.44 of the softmax mass on Qwen and 0.21 to 0.32 on Llama, against 0.90 to 0.95 without IDF. On every task with a mass reference, each configuration with IDF holds less mass than any configuration without (Figure 2, right).

As Proposition 1 predicts, saturation without IDF changes task scores almost nowhere. On Qwen, `010` and `000` differ in task score on 38 of 2,600 example–budget pairs and in token F1 to the full-attention output on 448, which we attribute to floating-point ties.

The registered frequency-curve predictor fails on both models. On Qwen it predicts a gain on all 24 rows and is wrong on every one, with an MAE of 4.18 against the limit of 1.81. On Llama it is wrong on every row with a nonzero margin, with an MAE of 4.00. Its predictions on Qwen lie between 2.13 and 2.21, a gain for nearly any input, so this failure is weak evidence about page frequency.

With exact attention mass as $\tilde{Q}$, the registered label-free rule chooses `010` on every row where the reference exists (26 on Qwen, 24 on Llama), since the pages chosen with IDF always hold less mass. On the rows with $G \neq 0$ (24 and 22) its sign is always right, and its choices reach the oracle's mean accuracy (0.742 over the 26 Qwen rows and 0.484 over the 24 Llama rows), against 0.024 and 0.001 for all-on. Always choosing `010` does exactly as well, so these rows cannot show that the rule adds anything over a constant. The registered test fails on the sign mark. No row has $G > 0$, so always predicting a loss is also right on every row, and the pass mark requires exceeding it. Across rows, $\tilde{G}$ and G have Spearman -0.32 on Qwen and -0.03 on Llama, so $\tilde{G}$ gets the direction but

Model		Budget	Full	Quest	H2O	000	001	010	011	100	101	110	111
Qwen	Accuracy	512	.811	.705	.181	.705	.719	.707	.718	.026	.015	.023	.022
		1024	.811	.778	.266	.778	.771	.777	.771	.036	.046	.026	.027
	Mass	512	1	.904	–	.904	.916	.904	.916	.327	.333	.291	.294
		1024	1	.942	–	.942	.950	.942	.950	.437	.443	.404	.407
Llama	Accuracy	512	.494	.433	.091	.433	.448	.433	.448	.000	.002	.002	.002
		1024	.494	.459	.136	.459	.465	.459	.465	.000	.000	.000	.000
	Mass	512	1	.896	–	.896	.912	.896	.912	.241	.245	.208	.210
		1024	1	.938	–	.938	.947	.938	.947	.310	.316	.272	.275

Table 3: Mean RULER accuracy (13 tasks, 4K context, 100 examples each) and mean share of full attention's softmax mass on the pages each configuration would select, at token budgets of 512 and 1,024. Full is full attention, and Quest equals 000 on every example. Llama's mass covers the 938 of 1,300 examples where full attention gives a non-empty answer.

not the size. Token F1 to full attention's output, the secondary reference, gives the same decisions.

7.2 Expert routing

In routing IDF also hurts, by less. Configuration 000 is the native forward pass. By Proposition 1, 001 to 011 give the same result as 000 on every example, and 101 the same as 100 (Table 9 in the appendix). With IDF on, mean MMLU accuracy falls from 0.480 to between 0.382 and 0.390, and WikiText perplexity rises from 7.49 to 7.86 or 7.88. All-on is worse than all-off on 52 subjects and better on 3, and significantly worse after Holm correction in 22 of the 58 rows (57 subjects and WikiText). IDF sends tokens towards experts the router rarely chose during calibration. The frequency bias loses less (MMLU 0.428, perplexity 7.73). The frequency-curve predictor predicts m of 2.05 to 2.18 for every subject, right in sign on 3 of the 55 subjects with a nonzero margin (0.05) and wrong on WikiText, and its MAE is 3.98 against the limit of 1.81. No label-free rule was registered for routing, because its planned reference, a mixture over all 64 experts, was not implemented.

8 Conclusion

Saturation. On sparse codes derived from E5, saturation gives most of BM25's gain without collection statistics, and cv_x ranks its effect at held-out values of k_1. Where each item carries a single score and IDF is off, saturation cannot change the selection up to ties (Proposition 1), and in KV selection and routing it changed task scores almost nowhere.

IDF and length normalisation. BM25 without IDF beats all-on at every support budget but 16/64. Whether IDF and length normalisation help a code on a collection is harder to say. The frequency curve does not predict it, and neither do index statistics. Agreement with text BM25 helps a fitted model, but the label-free rule built on it is no better than a constant in sign on new E5 rows and fails its registered test on SPLADE-v3, where agreement favours configurations that behave like BM25. Dense E5, the secondary reference, met both pass marks there, over only 10 collections and with an interval for its gain that includes zero.

Frequency inside the model. With frequency from the selector's own choices, IDF down-weights the items the selector picks often, and accuracy falls in KV selection on two models and in expert routing. In KV selection exact attention mass shows this without labels. The rule built on it chooses as well as the oracle, as does always choosing 010. No row has $G > 0$, so its registered sign test cannot be passed, and $\tilde{G}$ does not track the size of G.

Challenges, Limitations and Future Work

One encoder underlies every retrieval code, MIR-ACL is in E5's training data, and SPLADE is one model. The KV runs use one 4K context and 100 examples per task, not the planned 500. Variable tracking ran without RULER's answer prefix. Llama's full-attention answer is empty on 362 of 1,300 examples, all 100 of the first single-needle task among them, so its mass reference covers 938. Routing uses one model. A frequency counted over unrelated prompts is the next test.

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A Further results

Test (registered)	Data	Pass mark	Outcome
Frequency-curve predictor of m (19 Sep.)	179 retrieval rows	none; held-out R^2 reported	R^2 .04, .02 (basis families, corpora held out); sign .90, equal to always predicting a gain
Saturation mechanism, H1–H6 (29 Sep.)	k_1, b sweep	a threshold per hypothesis	all pass (Table 6)
Label-free rule, SPLADE-v3 (20 Oct.)	10 collections	sign above the better constant; rule $\geq$ all-on	fails (sign .50 with text BM25); the dense reference meets both (.70)
Frequency-curve predictor, KV (Qwen 20 Oct., Llama 23 Oct.)	24 and 23 RULER rows	sign $\geq$ .70, MAE ≤ 1.81, beat always predicting a gain	fails (sign .00 on both; MAE 4.18, 4.00)
Label-free rule, KV (same dates)	24 and 22 RULER rows	as for SPLADE-v3	fails (sign 1.00 equals the better constant); rule equals the oracle
Frequency-curve predictor, routing (20 Oct.)	WikiText, 57 MMLU subjects	as in KV	fails (sign .05, MAE 3.98)

Table 4: Tests fixed in a dated commit before their results were read. For SPLADE-v3 the scores were already known. We added the Llama tests after reading the Qwen results and before any Llama run, noting in advance that the KV sign mark cannot be met if no row has $G > 0$; we had not foreseen this for Qwen. Sign is the fraction of rows with a nonzero target whose sign is predicted correctly. The label-free rule of our interim report, evaluated on MS MARCO rows produced afterwards, is reported in Section 5.5 but was not registered.

Shuffled and uniform frequencies. On ArguAna, NFCorpus and SciFact we replaced the real frequencies in two ways, over 42 rows each. With shuffled frequencies IDF's mean Shapley value turns negative, and IDF's Shapley value with the real frequencies is larger in 83% of rows (one-sided Wilcoxon $p < 10^{-6}$). Uniform frequencies remove IDF altogether, and all-on nDCG@10 is then 0.157 against 0.147 with the real frequencies ($p = 0.15$).

SPLADE-v3 by configuration. Configuration `110` scores 0.528 on SPLADE-v3, and adding length normalisation (`111`) gives 0.509, a drop of 0.019. Length normalisation's Shapley value, −0.073 (Table 2), is larger in magnitude because it also averages over `001` and `101`. These divide by length without saturation and score 0.13 below `000` and 0.12 below `100`. Saturation's Shapley value of 0.048 comes from adding it to `001` and `101`. Added to `000` or `100`, it changes mean nDCG@10 by less than 0.01.

Basis	Gran.	`111`	cv_x	ϕ_{sat}	IDF + len.	$G > 0$
identity	pooled	.148	0.36	.044	.68	1.00
	token	.142	0.57	.061	.55	.77
Gaussian	pooled	.098	0.28	.020	.77	.85
	token	.088	0.51	.030	.64	.92
PCA	pooled	.221	0.71	.138	.24	.69
	token	.310	1.22	.155	.45	.46
k-means	pooled	.018	0.05	.000	–	.31
	token	.185	1.65	.125	.11	.85
PQ	pooled	.012	0.00	.000	1.00	.62
	token	.144	1.36	.099	.09	.69
RQ	pooled	.056	0.00	.000	1.00	.92
	token	.292	1.97	.176	.18	.69
SAE	pooled	.127	0.18	.014	.66	.77
	token	.375	3.02	.193	.32	.46

Table 5: Per code (Gran. is granularity), over its 13 primary rows, the mean all-on nDCG@10, mean cv_x, mean saturation Shapley value, share of the gain from IDF and length, and fraction of collections with $G > 0$. Pooled k-means has no gain to share.

	Hypothesis	Result
H1	IDF + length share at default in [.36, .42]	.393
H2	ϕ_{sat} at $k_1 = 0.6$ exceeds $k_1 = 2.4$	.093 > .080
H3	within-corpus $\rho(\mathrm{cv}_x, \phi_{\mathrm{sat}}) \geq 0.3$	.89, .89
H4	cv_x model ranks ϕ_{sat} at $\rho \geq 0.5$	.87, .90
H5	$\phi_{\mathrm{len}} = 0$ exactly at $b = 0$	max 0.0
H6	ϕ_{len} at $b = 0.75$ exceeds $b = 0.5$	.045 > .029

Table 6: Confirmatory tests, committed before they were run, on the k_1 and b sweep. ρ is Spearman, and the default is $k_1 = 1.2$, $b = 0.75$. H2 and H6 have Holm-corrected $p = 3.6 \times 10^{-5}$. H3 and H4 are tested at $k_1 = 0.6$ and 2.4; H4's model is fitted at $k_1 = 1.2$. H5 covers 210 rows. H1 uses the default rows of the five sweep collections.

Sweep	Coll.	Rows
primary, 14 codes	13	182
support budgets 16/64, 64/256	13	364
native support	3	42
$k_1 \in \{.6, 1.2, 2.4\}$, $b \in \{0, .5, .75\}$	5	560
basis seeds 1 and 2	5	100
shuffled, uniform *df* controls	3	84
shared SAE dictionary	13	26
all rows	13	1,358

Table 7: Retrieval rows, each with eight configurations. Rows a sweep shares with the primary rows are counted once, so the rows sum to the total. Native support is uncapped, and the shared SAE dictionary applies the token-level SAE dictionary at both granularities.

Target	Features	Family	Corpus	Params
ϕ_{sat}	frequency curve	$-.02$	.16	.06
	index	.33	.29	.56
	pseudo-labels	.66	.37	.77
ϕ_{len}	frequency curve	$-.30$	$-.25$	.01
	index	$-.13$	.25	$-.09$
	pseudo-labels	.59	.68	.82
ϕ_{idf}	frequency curve	$-.07$	$-.05$	.02
	index	$-.05$	.04	.08
	pseudo-labels	.37	.33	.76
G	frequency curve	.01	.07	.13
	index	.01	.03	.12
	pseudo-labels	.53	.44	.59
	pseudo-labels, E5	.63	.56	.78

Table 8: Held-out R^2 of exploratory ridge regression on the 181 primary rows whose features are all defined, holding out basis families or corpora, or predicting the 560 rows at other k_1 and b. The index tier adds index statistics to the frequency-curve features, and the pseudo-label tier adds ranking features and pseudo-qualities. Pseudo-labels use text BM25 unless marked E5 (the dense reference).

Routing	MMLU acc.	WikiText ppl.
native, `000`–`011`	.480	7.49
frequency bias	.428	7.73
`100`, `101`	.390	7.86
`110`	.386	7.88
`111`	.382	7.88

Table 9: Mean zero-shot accuracy of OLMoE-1B-7B over the 57 MMLU subjects (14,042 questions) and WikiText-2 perplexity, for the native router, the load-driven frequency bias and the eight configurations. Configurations in one row route identically.